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SARDAR PATEL UNIVERSITY

M. Sc. Integrated Biotechnology (IGBT) 7th Semester

Theory Examination - November 2021

PS07EIBI22 – Microbial technology

26th November 2021 (Friday), 12:00 pm to 02:00 pm

26/11/2021, Friday, 12:00 PM

Total Marks: 70

Note: (1) All the Questions are compulsory.
(2) Figures on the right indicate marks.

Q.1 (A) Choose the correct option

1 × 8 = 08

- (i) Who among the following is **not** credited for the discovery of Streptomycin?
[A] E. Bugie [B] A. Schatz [C] S. Waksman [D] A. Fleming
- (ii) Xanthan is not made up of _____ sugar/s?
[A] Glucose [B] Mannose [C] Xylose [D] Glucuronic acid
- (iii) Direct carboxylation of pyruvate catalyzed by malic enzymes provides malate which is readily oxidized into oxaloacetate through _____.
[A] Pyruvate carboxylase [B] Pyruvate dehydrogenase
[C] PEP decarboxylase [D] Malic dehydrogenase
- (iv) Which of the following examples are **not** related with the class of metabolite, that is "Products directly associated with generation of energy in the cell"?
[A] Lactic acid [B] Penicillin [C] Ethanol [D] Butanol
- (v) The $\Delta 1$ -dehydrogenation of hydrocortisone to prednisolone by *Streptomyces affinis* using enzyme _____.
[A] synthetase [B] hydrolase [C] dehydrogenase [D] All of these
- (vi) If starch containing substrates are used for ethanol production, yeast strain can't be used directly because _____.
[A] it is converted to pentose sugar
[B] starch is not a suitable substrate to produce ethanol
[C] it doesn't contain amylases to hydrolyze starch
[D] none of these
- (vii) The Malolactic fermentation undertaken by lactic acid bacteria (LAB) in wine fermentation is **not** due to _____.
[A] *Acetobacter aceti* [B] *Oenococcus oeni*
[C] *Pedicoccus damnosus* [D] *Lactobacillus plantarum*
- (viii) The first commercial SCP "PRUTEEN" *Methylophilus methylotrophus* was produced by ICI using _____.
[A] Ethanol [B] Molasses [C] methanol [D] None of these

Q-1 (B) Do as directed

1 × 16 = 16

Fill in the blanks

- 1) _____ is the production and study of grapes which deals with series of events that occurred in vineyard.
- 2) Microorganisms produce secondary metabolites like antibiotics usually during _____ phase of growth.
- 3) 2-Hydroxypropane-1,2,3-tricarboxylic acid is the chemical name of _____.
- 4) Addition of _____ precursor in fermentation media, to produce Penicillin V by *Penicillium chrysogenum*.
- 5) Starch is saccharified using _____ enzyme whose production can be done using of solid-state fermentation (SSF).
- 6) During citric acid fermentation the Copper is used at up to 500 ppm as an antagonist of the enzyme _____ which requires iron.
- 7) _____ act as a precursor in Vitamin B₁₂ biosynthesis, which is, converted δ-amino levulinic acid in presence of glycine.
- 8) Vegetable oils e.g., corn oil or soybean oil is added in the fermentation medium for antibiotic production, act as _____.

State True or False:

- 9) Ergot alkaloids are formed in sphacelial stage.
- 10) Papain is the best-known cysteine protease.
- 11) Surfactants are surface active agents, reducing the surface tension in the foams and destabilizing protein films.
- 12) Strict sterile conditions are not required in antibiotics production because the antibiotic producing organisms can be cultivated under unsterile conditions.
- 13) Alexander Fleming, Ernst Chain and H.W. Florey are the Scientists who got Nobel prize for the discovery of Penicillin.
- 14) Lactic acid bacteria use phosphoketolase pathway for heterofermentation of glucose.
- 15) Crude media or complex media provides the fewest problems while downstream processing.
- 16) Bitter Ale is an originally top-fermented beer.

Q-2 Attempt any seven of the following:

2 × 7 = 14

- i. What is amylose and amylopectin.
- ii. Explain the effect of biotin on glutamic acid fermentation.
- iii. Name any two raw substrates and their sources which are used for the synthesis of steroid drugs.
- iv. Explain the biological functions of microbial exopolysaccharides.
- v. Write the desirable characteristics of organism to be used as SCP.
- vi. Enlist the applications of proteases
- vii. Explain the mode of action of Streptomycin.
- viii. Write the 2 names of rennin like enzyme producing microorganisms.
- ix. List the biochemical functions of cyanocobalamin.

Q-3 Discuss the fermentative production of industrial ethanol in detail. [08]

OR

Q-3 Explain in detail the fermentative production of Amylase. [08]

Q-4 Explain in detail downstream processing in penicillin fermentation. [08]

OR

Q-4 Explain in detail on saprophytic production of ergot alkaloids. [08]

Q-5 What is SCP? Write in detail on production and properties of bacterial SCP. [08]

OR

Q-5 Write in detail on cultivation of edible and medicinal mushrooms. [08]

Q-6 Neatly narrate flow diagram for cheese making & explain the coagulation of milk during cheese making. [08]

OR

Q-6 Explain in detail the malting and mashing process in beer production. [08]

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